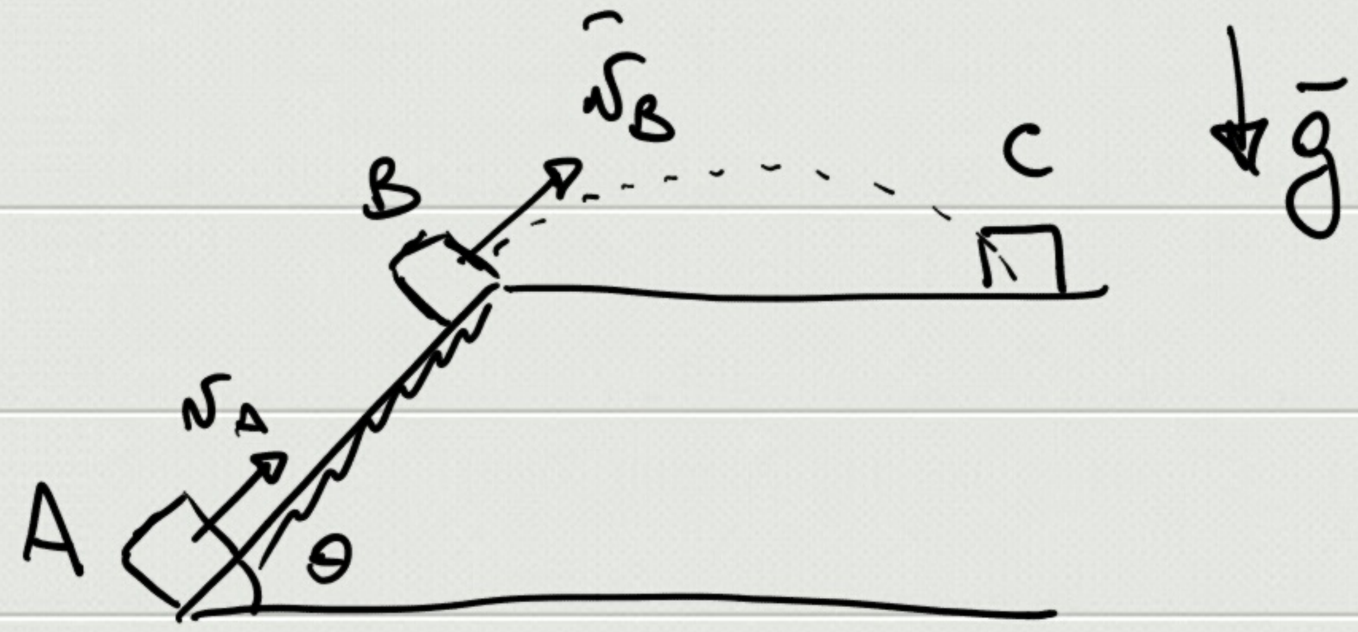


$\theta = 40°$

$\mu_d = 0.4$

$AB = BC = d = 1.43\,m$

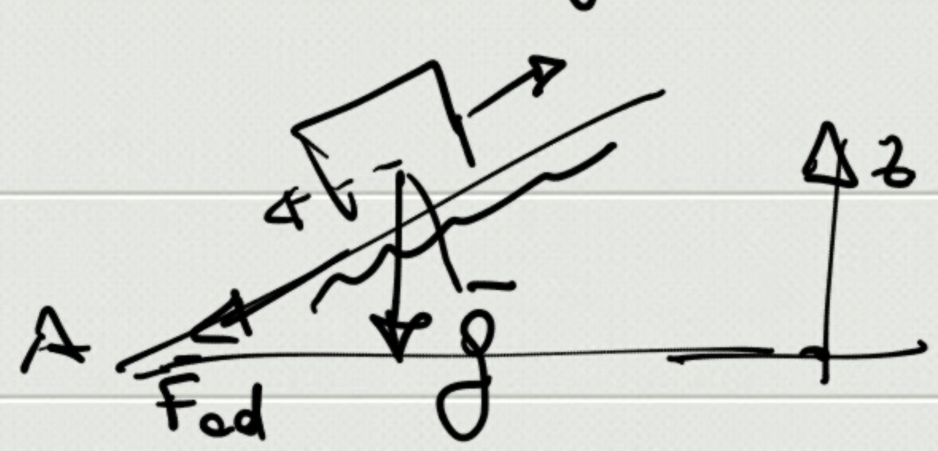
$v_A = ?$

$$v_B^2 = v_A^2 - \cancel{2\mu_d g}\cos\theta \cdot d$$

$$v_B^2 = v_A^2 - 2(\mu_d g\cos\theta + g\sin\theta)d \quad *$$

$$* \quad -\mu_d \cancel{m} g\cos\theta \cdot d = \left(\frac{1}{2}\cancel{m} v_B^2 + \cancel{m} g d \sin\theta\right) - \frac{1}{2}\cancel{m} v_A^2 \quad *$$

$$\frac{1}{2} m v_B^2 \cancel{= \frac{1}{2} m v_A^2} - m g d \sin\theta$$



$$v_f^2 = v_i^2 + 2al$$

$$W_{mc} = \Delta E_m = \Delta(E_k + E_p)$$

$$\int_A^B \bar{F}_{ad}\, d\bar{s} = -\mu_d N \int_A^B ds = -\mu_d mg\cos\theta \cdot d$$

$$\boxed{v_B^2 = v_A^2 - gd(\mu_d\cos\theta + \sin\theta)}$$

$$\Rightarrow v_A^2 = v_B^2 + gd(\mu_d\cos\theta + \sin\theta)$$

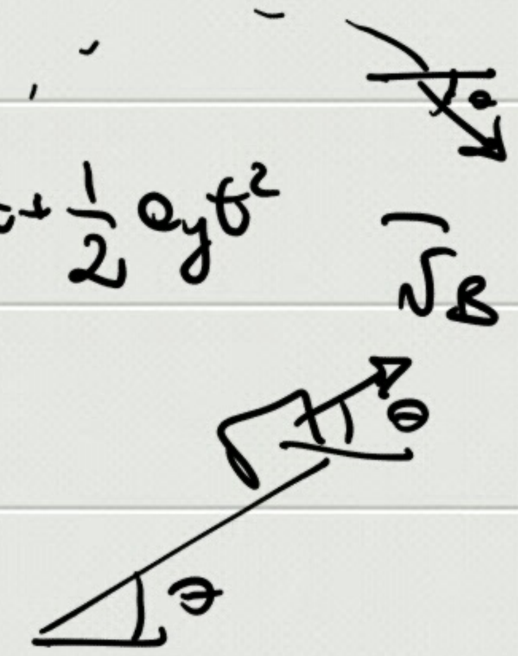
$$* - \begin{cases} \underbrace{v_B \cos\theta} \cdot t = d \\ -v_B \sin\theta = v_B \sin\theta - g t \end{cases}$$

$$x(t) = \cancel{x_0} + \underbrace{v_x} t$$

$$v(t) = v_0 + a t$$

$$* - \begin{cases} v_B \cos\theta \cdot t = d \\ 0 = v_B \sin\theta \cdot t - \frac{1}{2} g t^2 \end{cases}$$

$$y(t) = y_0 + v_{0y} t + \frac{1}{2} a_y t^2$$



$$- \quad 0 = v_B^2 - 2 g d$$

$$- \quad 0 = \frac{1}{2} m v_B^2 - m g h$$

$$d = \frac{2 v_B^2}{g} \sin\theta \cos\theta \quad \Rightarrow v_B = \ldots$$

$$t = \frac{d}{v_B \cos\theta}$$

$$g t = 2 v_B \sin\theta$$

$$\boxed{v_B^2 = \frac{g}{2 \sin\theta} \frac{d}{\cos\theta}}$$

$$v_A = \sqrt{d g \left[\frac{1}{2 \sin\theta \cos\theta} + 2(\sin\theta + \mu \cos\theta) \right]} =$$

$$= 6.4 \ m/s$$